Ben Barlow

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Key: Grade | Hyperlink | Skill

Education

University of Edinburgh · MSc in Artificial Intelligence — *Edinburgh, UK*
Sep 2022 - Aug 2024

Graduated with Distinction (maximal classification) | Dissertation: 84%

Machine Learning Pattern Recognition · Probabilistic ML · Reinforcement Learning · Deep Learning | PyTorch | Tensorflow

Stats Bayesian Statistical Models · Monte Carlo Methods · Missing Data & Imputation | JAGS | Stan

University of Warwick · BSc in Computer Science & Statistics — *Coventry, UK*
Oct 2017 - Jun 2020

Graduated with First-Class Honours (maximal classification) | Dissertation: 84%; Class rank: 1st

Stats Parameter & Interval Est. · Hypothesis Testing · Freq. Stat. Models | **Maths** Linear Algebra · Multivar. Calculus · Real Analysis

CS OOP · Algorithms · Data Structures · Databases · Software Engineering | **Probability** Foundations · Stochastic Processes

Work Experience

Rokos Capital Management (via Neptune North) — Macro hedge fund, $20bn AUM — *Newcastle & London, UK*

Data Scientist & Engineer | Python | SQL | Azure Databricks — *Apr 2025 - Present*

Analyse and ingest alternative data from financial vendors to generate trading edge, liaising closely with Quants and Economists.

- *Current Project*: Building inflation forecasting models on Infare data by mirroring index methods used by statistical authorities, addressing time series challenges (seasonality, anomalies during COVID) and evaluating predictive performance against benchmark models.
- Deployed a data analytics tool for time series data quality checks, applying statistical techniques for outlier detection on market data.
- Pioneered adoption of Databricks apps by evaluating tools (e.g., Plotly Dash, OLTP Databases) aligned with low-latency dashboard needs.
- Implemented Databricks Asset Bundle deployment of apps to allow software engineering best practices (source control, unit testing, CI/CD).
- Proactively pursued CFA Investment Foundations and Databricks Certified Machine Learning Associate certificates.
- Trained peers in writing clean, modular Python code, the PEP8 guidelines, and using code quality tools such as `ruff`.

School of Informatics (University of Edinburgh) & FC Laboratories Ltd — *Edinburgh, UK*

Python Developer & Data Scientist | Python | Poetry | matplotlib | networkX | pytest | Git — *Jan 2023 - Jan 2025*

- Balanced academic studies and part-time jobs in data science, working in a university research environment and an early-stage startup.
- Conceptualised, implemented, and unit tested Python packages, addressing performance bottlenecks through memory and time profiling.

Account Technologies — FinTech, credit provider, £100mn revenue, 500k customers — *London, UK*

Decision Scientist | Python | SQL | pandas | numpy | plotly | pytest | Airflow | Docker | SQLAlchemy — *Oct 2020 - Jul 2022*

Interacted with transactional, credit bureau, operational, financial, compliance, marketing, and customer data using Python to develop bespoke packages (deployed via CI/CD pipelines) and the PyData stack to visualise data, train machine learning models, and report on business metrics.

- *Example Project 1*: Increased profitability (£100,000s) by identifying populations of unprofitable customers and blocking their acquisition. Found populations using decision trees split on consumer spending habits with automated search over hyperparameter space. Tested statistical significance of populations through single-group z-tests. | Airflow | Hypothesis Testing | Hyperparameter Tuning
- *Example Project 2*: Created a new credit limit increase strategy (logistic regression model) to efficiently allocate credit line to the business's most profitable customers. Conducted EDA, feature engineering, predictor and dependent variable selection, imputation for missing data, and model training & evaluation. Model uplift from 29p to 33p of profit per £1 credit limit increase. | EDA | pandas | plotly | SQL | scikit-learn

Research Experience

Intelligent Systems Lab & Mineral-X, Stanford University — Labs in AI & mineral exploration — *Palo Alto, USA*

Student Researcher (MSc Dissertation) | Advisor: Robert Moss | Blog | Code | Thesis — *Jun 2024 - Aug 2024*

Thesis titled 'Intelligent Optimisation of Borehole Geophysics Planning for Critical Mineral Exploration'. | Achieved 84%

- Received three weeks of in-person supervision from advisor (a Stanford computer science PhD candidate) and colleagues at his lab (SISL).
- Implemented path planning for an agent exploring a mineral exploration environment. | Julia | MCTS | (PO)MDP | Probabilistic Modelling
- Learned rapidly, as shown by integrating a new domain, AI technique, and programming language, delivering results within twelve weeks.

University of Warwick — *Coventry, UK*

Student Researcher (BSc Dissertation) | Advisor: Ioannis Kosmidis | Code | Thesis — *Oct 2019 - May 2020*

Thesis titled 'Shrinkage-driven Unsupervised Learning with Clustering Applications'. | Achieved 84% | Class rank: 1st

- Defined a new family of regularised copula-based mixture models that overcome the limitations of Gaussian mixtures. | Copula
- Implemented methods for model estimation (using ECM algorithm) and model selection (using silhouette width and BIC). | R | C++ | EM Alg.

Extracurricular Activity

University of Warwick Men's Football Club — *Coventry, UK*

Director of Football & First Team Captain — *Mar 2019 - Mar 2020*